
London Fire Brigade

Vehicle mobilisation and arrival times

Clemente Ibarra and Matías Muñoz Hoffmann

Business Intelligence C1

29 September 2026



Our question and decision



Which districts and dispatch hours have longer attendance times?



Decision: prioritise areas and hours for further investigation.



The public data supports quality checks, EDA and a possible C2 extension.

Data source and observation



Official source:
London Fire Brigade
Mobilisation Records.



One row represents
one vehicle mobilisation.

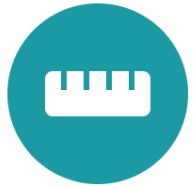


Analysis window:
dispatch dates in 2023–2024,
using GMT.



Several vehicles can
attend the same incident.

Our KPI: the 90th percentile



Measure: minutes from mobilisation to arrival.



P90: about 90% of eligible times fall at or below this value.

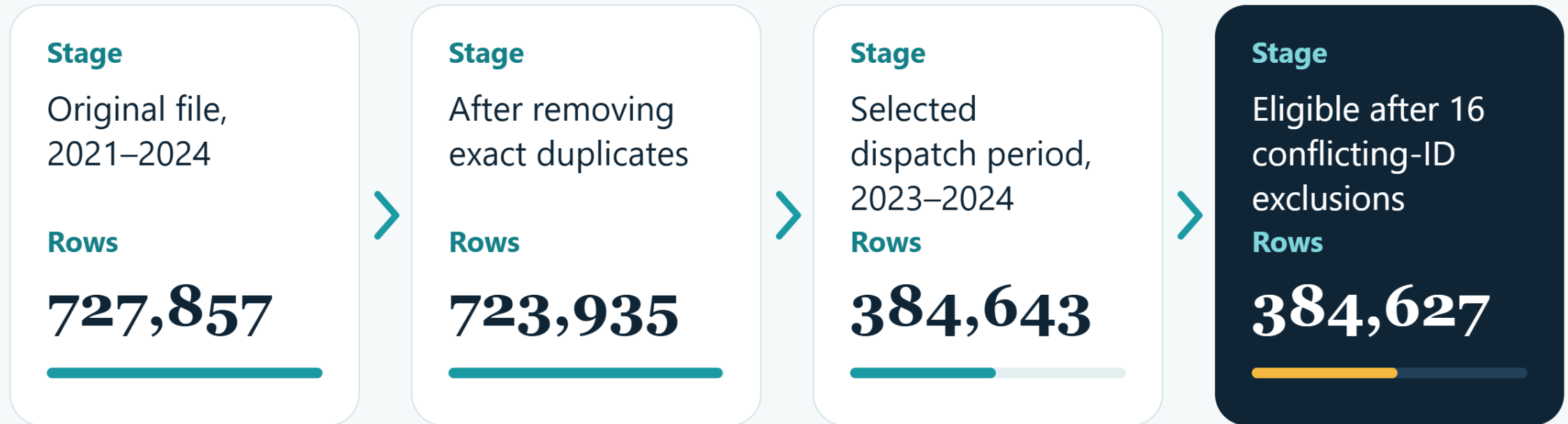


Purpose: examine longer times alongside the median and record count.



The measure excludes call handling before mobilisation.

Data preparation and row counts



Final population: 384,627 vehicle mobilisations across 248,225 distinct incident IDs.

Cleaning decisions and validation



Check unique mobilisation IDs and nonnegative durations.



Compare attendance time with arrival minus mobilisation, within **1 second**.

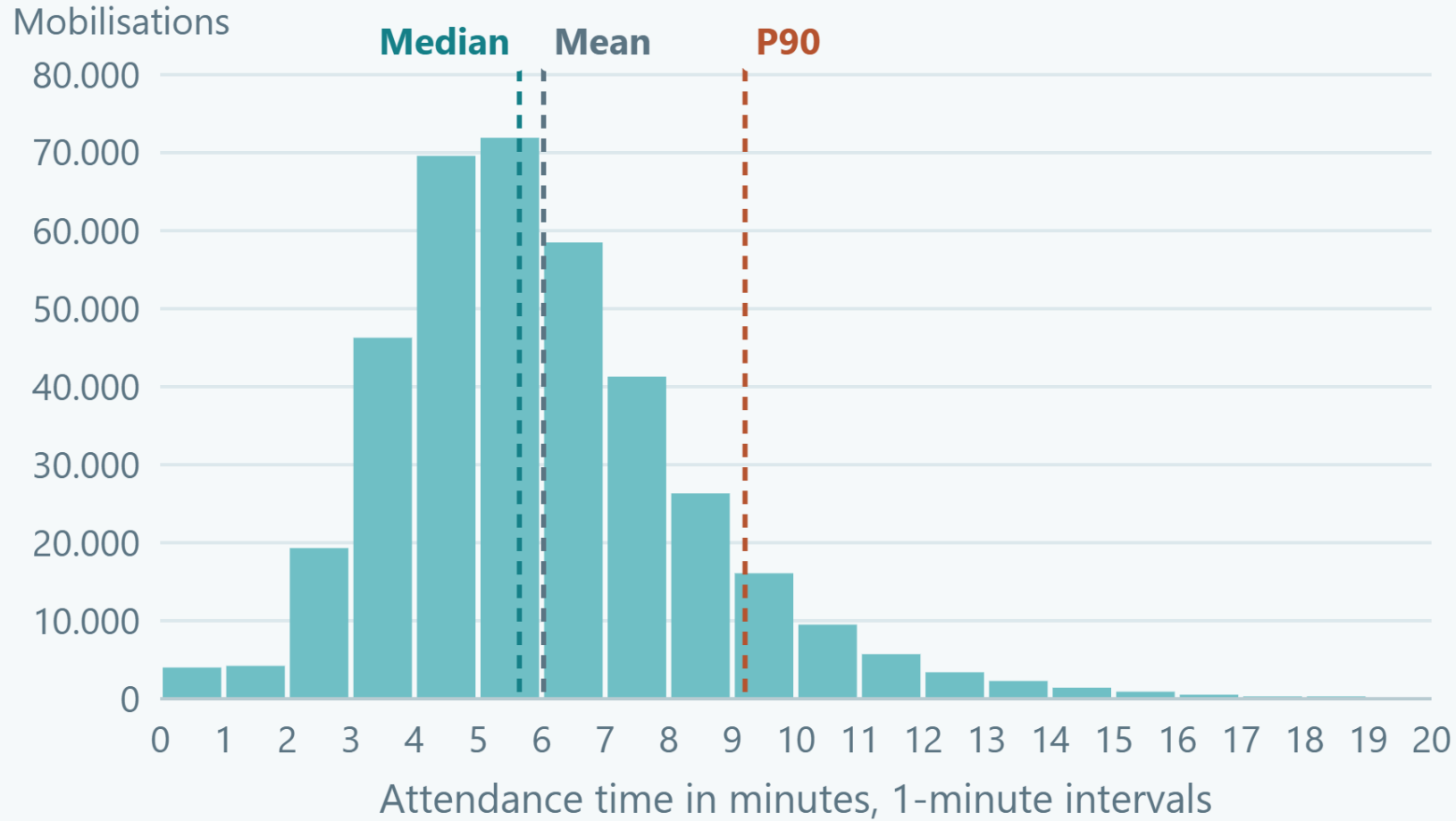


Keep **1,288** records with unknown district in the global KPI.



Keep valid long times. **Do not impute the target.**

Distribution of attendance times



Median

5.65 min

Mean

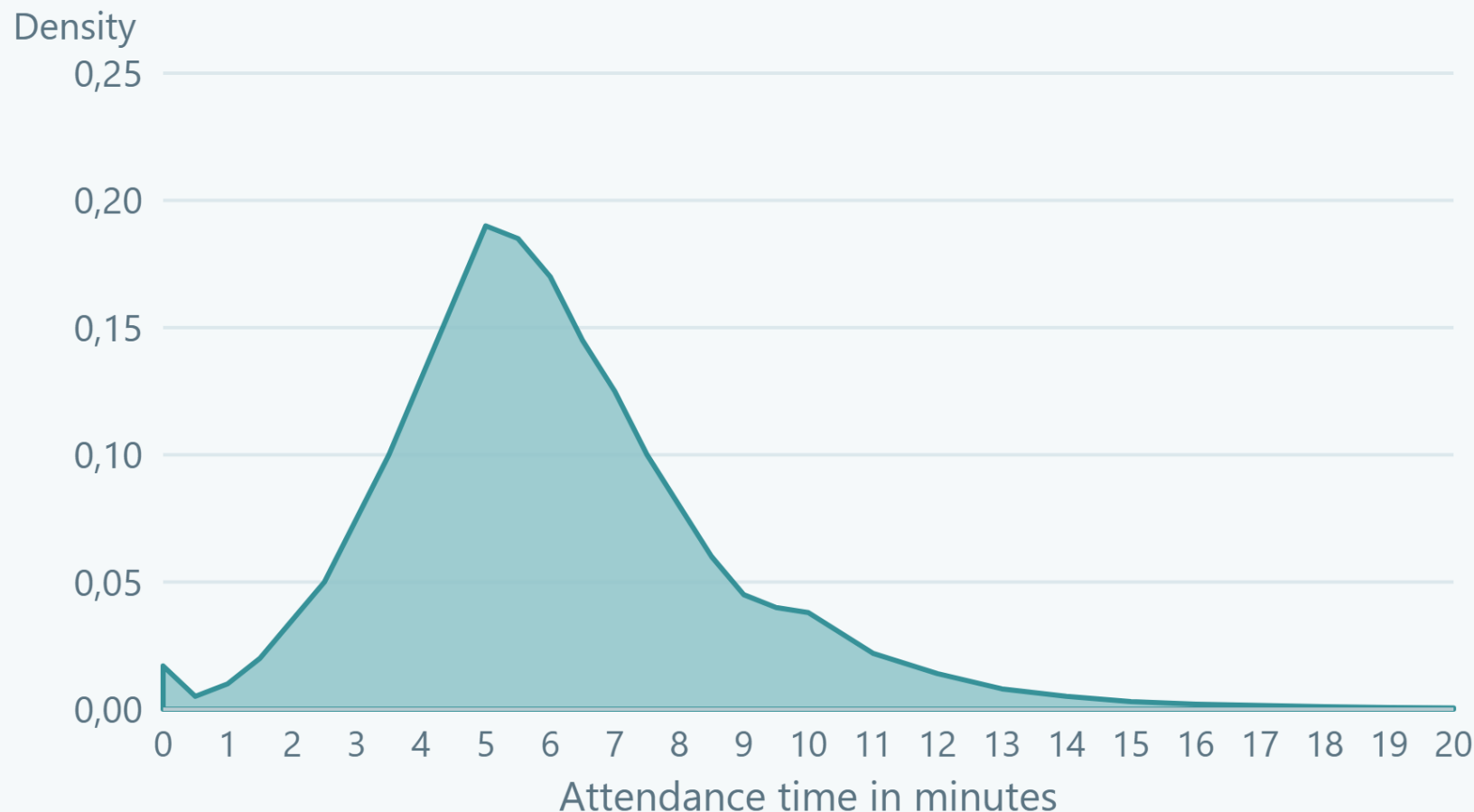
6.03 min

P90

9.20 min

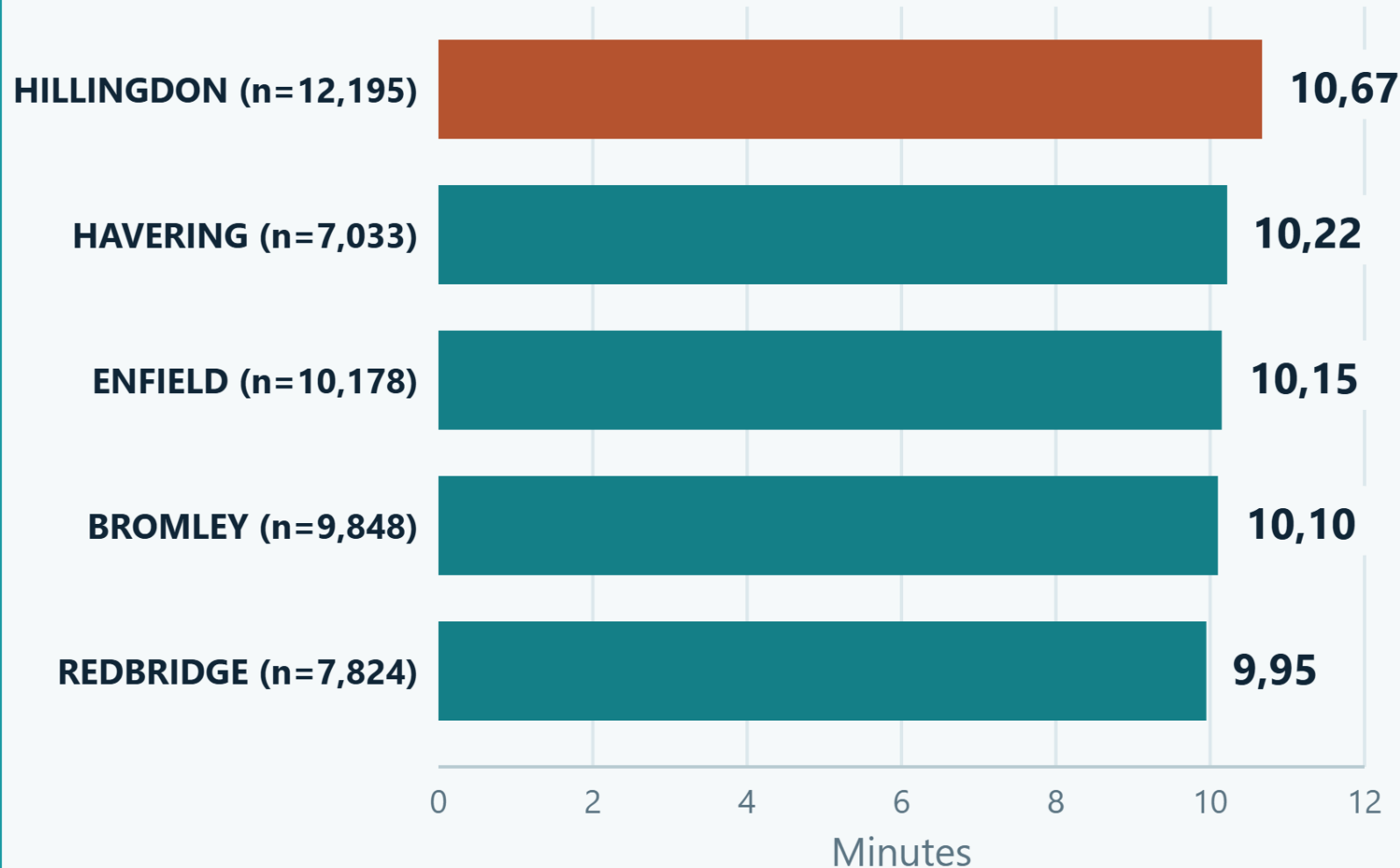
All 384,627 eligible records. Full observed range shown.

Density of attendance times



KDE based on a reproducible sample of 20,000 records, with reflection at zero. Curve height shows density. It is not a direct percentage.

Districts with the highest P90



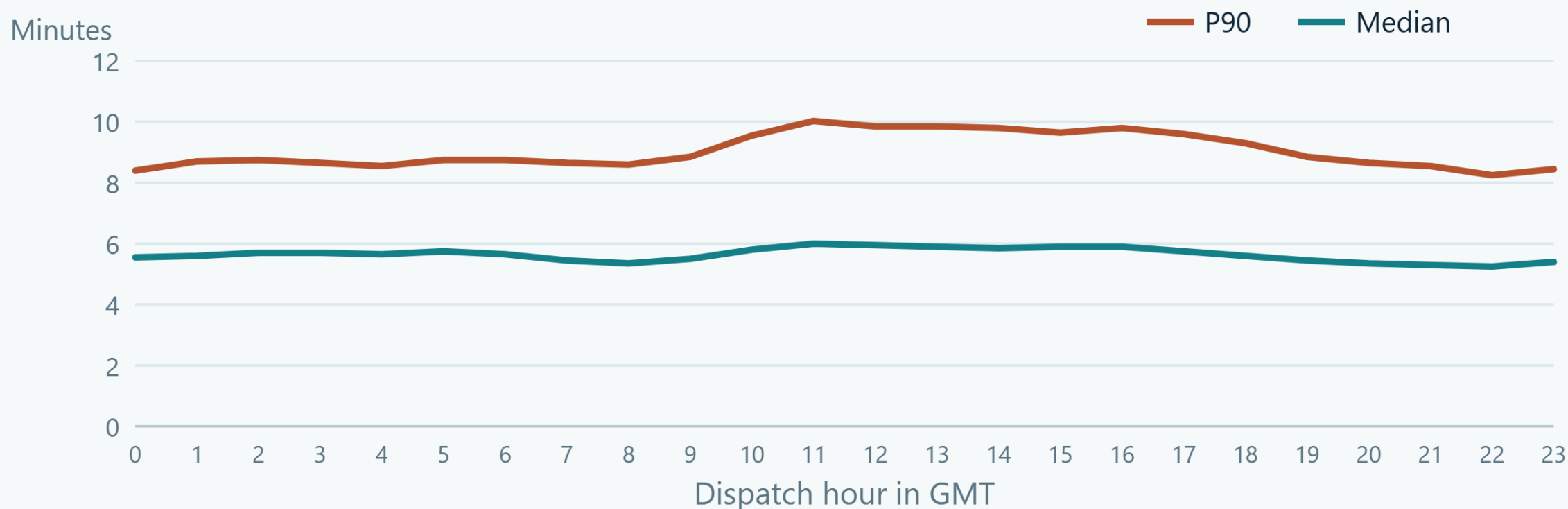
Hillingdon: P90 of 10.67 minutes, based on 12,195 eligible mobilisations.



Comparisons do not adjust for distance or incident mix.

Attendance times by dispatch hour

Highest hourly P90: 11:00–11:59 GMT, approximately 10.03 minutes.



Each group combines all available dates in the selected period.

Correlation between time components

Spearman correlation

Coefficient

0.952

Travel time and total attendance time

Coefficient

0.169

Turnout time and total attendance time

Coefficient

-0.083

Turnout time and travel time



Total time includes turnout and travel.
Correlation does not measure a share of total time.



Complete, consistent component records:
382,811. No causal claim.

A rule for further review



Monthly district P90 above the same-month global P90



At least 100 eligible records per month



At least 3 consecutive months

19 districts

with 3 consecutive months

15 districts

with 6 consecutive months



Sensitivity period: January 2023–November 2024. Counts unchanged for $n \geq 50$, 100 or 200. Exploratory rule proposed by the project. No official service threshold.



Limits of the evidence



No records for 31 December 2024.



All observed records are Initial. Durations stop at 20 minutes.



The data does not control for distance, incident type or resource availability.



Several vehicle records can belong to one incident.

Conclusion and next step



Prioritise districts and hours for investigation.



Check local context before changing resources.



C2: compare a baseline with models using information available at dispatch.



Use C1 feedback to improve the next stage.

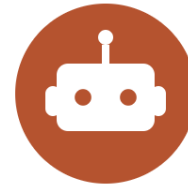
Reproducibility and AI use



GitHub contains the code, notebook, results and documentation.



The source manifest records download details and file hashes.



AI supported the code and analytical preparation.



Each student remains responsible for the submitted decisions and claims.

Backup: measurement and code

`pd.to_datetime()` :

convert text into dates and times.

`drop_duplicates()` :

remove repeated complete rows.

`groupby()` :

calculate results separately for each group.

`quantile(.9)` :

calculate P90.

assert:

stop if a required check fails.

Backup: yearly summaries

Year

2023

Eligible records

186,339

Median (min)

5.62

P90 (min)

9.17

Year

2024

Eligible records

198,288

Median (min)

5.68

P90 (min)

9.25



2024 lacks 31 December. Small descriptive differences do not establish causes.

Backup: sources and project files



London Fire Brigade Mobilisation Records, London Datastore.



Source metadata and SHA256: data/source_manifest.json.



Methods: scripts/analysis.py and docs/KPI.md.



Evidence: outputs/*_metrics.csv, audit.json and summary.json.